GITAM Deemed to be University

Computer Science and Engineering | Course Code: CS101

Object-Oriented Programming & Systems - Object-Oriented Programming Princip

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Overview: Virtual tables vtable, dynamic dispatch, smart pointers, resource acquisition is init

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Paradigms & Object Concepts
Procedural vs Object-Oriented paradigms, Classes, Data encapsulation, Access specifiers, Constructor

Unit 2: Inheritance & Polymorphism
Single, multiple, and hierarchical inheritance, Diamond problem, Virtual base classes, Virtual funct

Unit 3: Memory Management & Generics
Stack vs Heap memory, Dynamic memory allocation, RAII idiom, Smart pointers unique_ptr, shared_ptr,

Unit 4: Exception Handling & File I/O Streams
Try-catch-throw mechanisms, Custom exception hierarchies, Stream classes, Formatted disk file persis

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Illustrate with memory diagrams how virtual method tables vtable and virtual pointers vptr achieve dyn
2. Describe the diamond problem in C++ multiple inheritance and show how virtual base classes resolve mul
3. Explain RAII Resource Acquisition Is Initialization and contrast unique_ptr with shared_ptr in referen

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Bjarne Stroustrup - The C++ Programming Language Addison-Wesley 4th Ed
* Robert Lafore - Object-Oriented Programming in C++ Sams Publishing

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